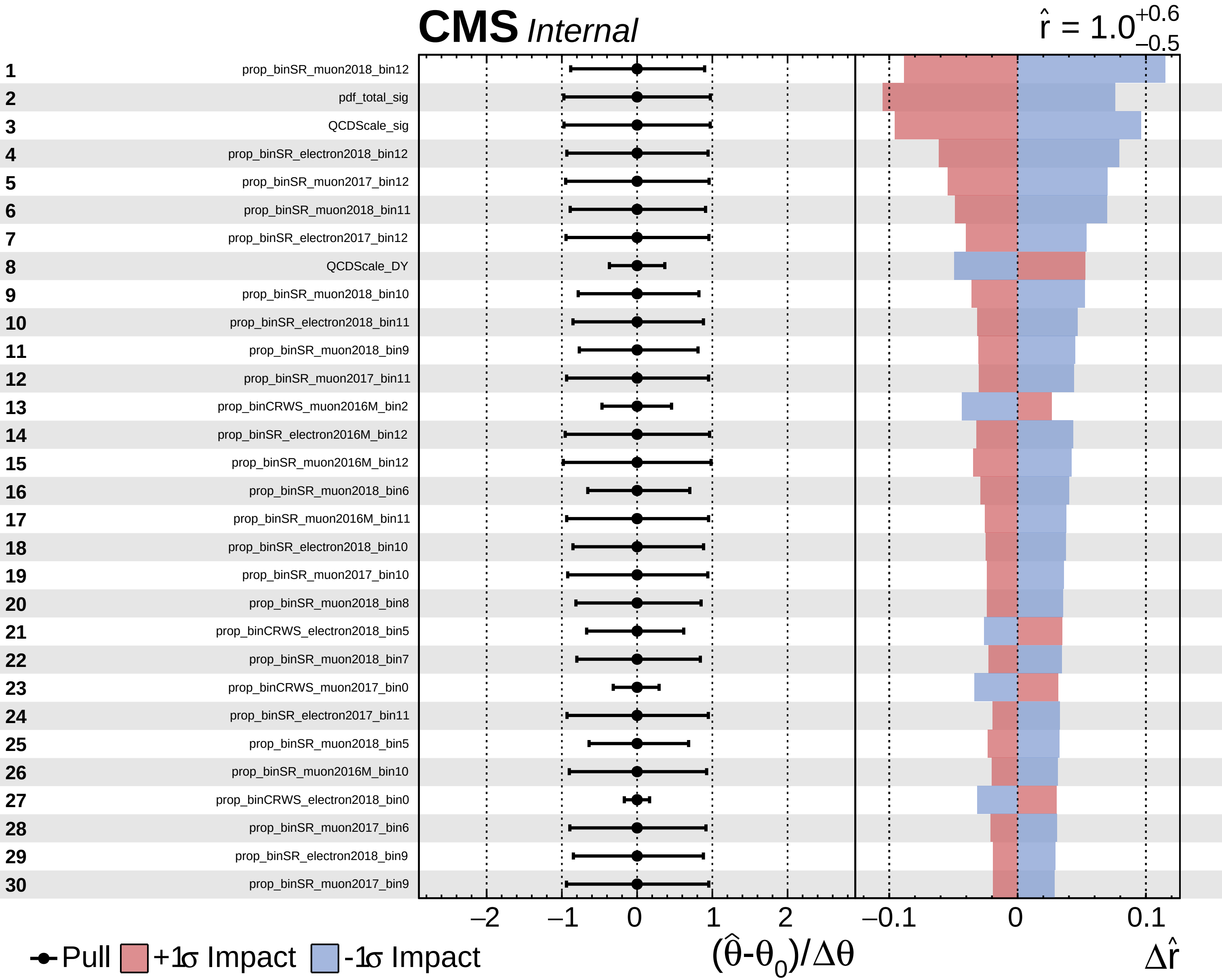
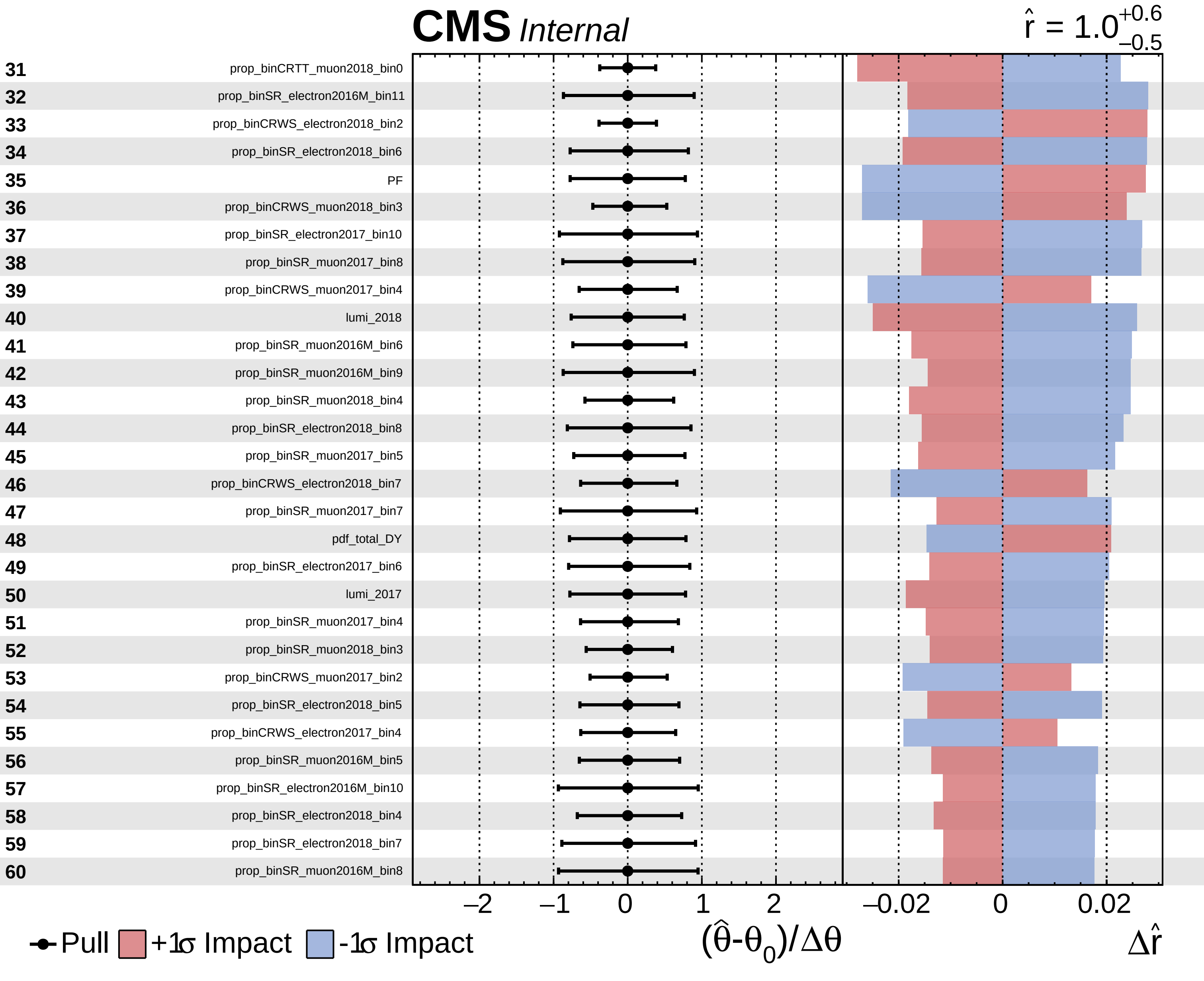
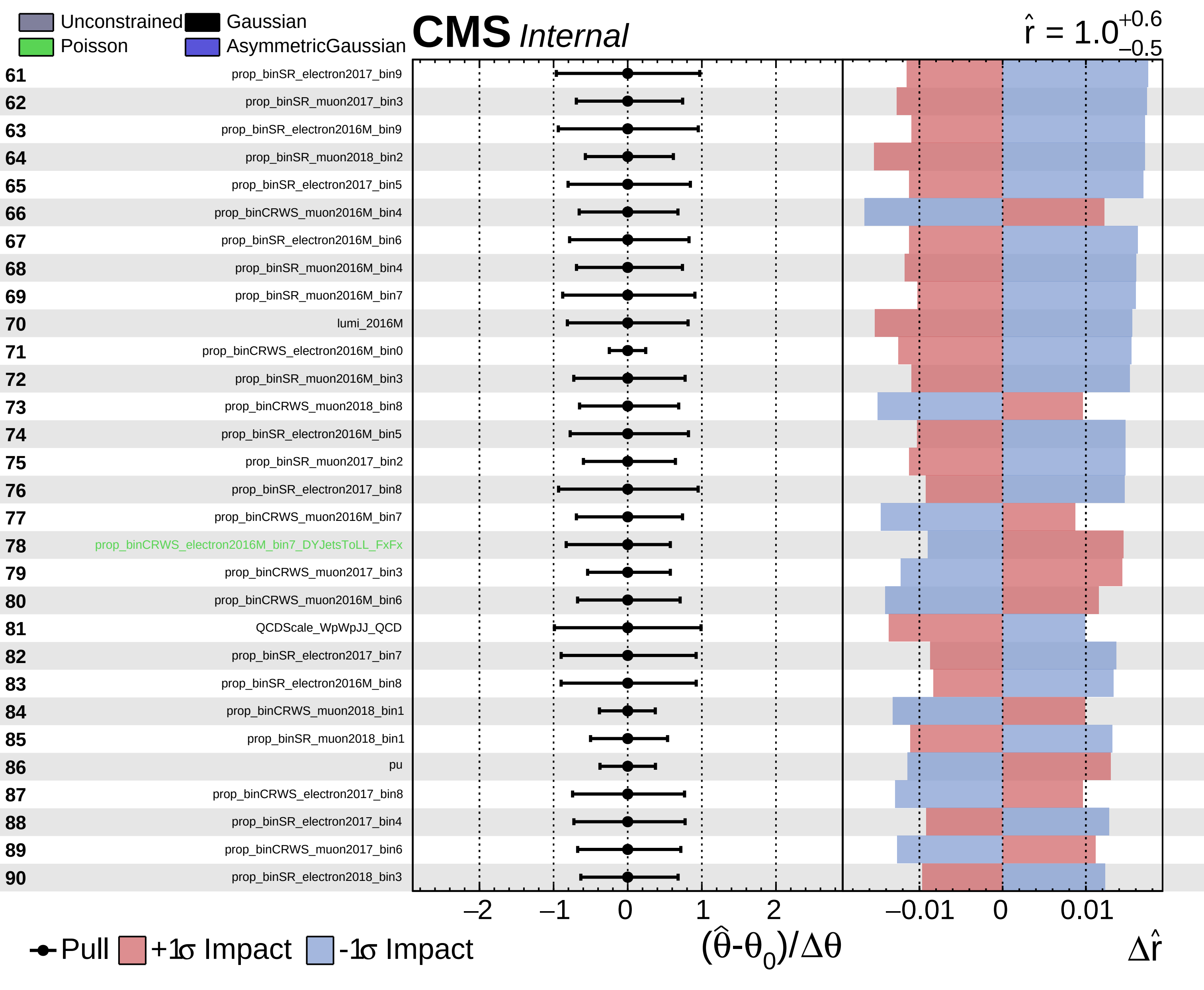
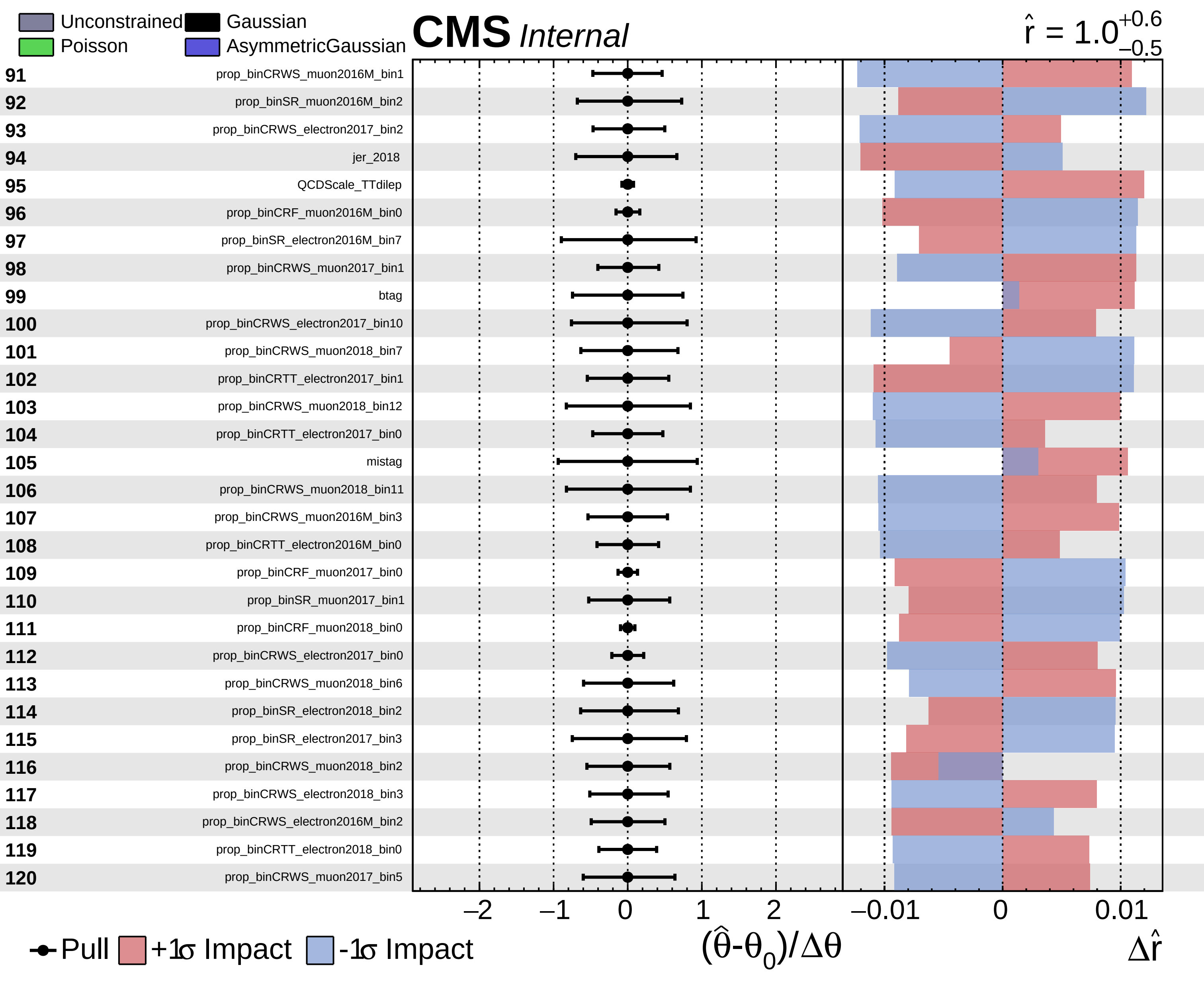


CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin12
2 pdf_total_sig
3 QCDScale_sig
4 prop_binSR_electron2018_bin12
5 prop_binSR_muon2017_bin12
6 prop_binSR_muon2018_bin11
7 prop_binSR_electron2017_bin12
8 QCDScale_DY
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2018_bin9
12 prop_binSR_muon2017_bin11
13 prop_binCRWS_muon2016M_bin2
14 prop_binSR_electron2016M_bin12
15 prop_binSR_muon2016M_bin12
16 prop_binSR_muon2018_bin6
17 prop_binSR_muon2016M_bin11
18 prop_binSR_electron2018_bin10
19 prop_binSR_muon2017_bin10
20 prop_binSR_muon2018_bin8
21 prop_binCRWS_electron2018_bin5
22 prop_binSR_muon2018_bin7
23 prop_binCRWS_muon2017_bin0
24 prop_binSR_electron2017_bin11
25 prop_binSR_muon2018_bin5
26 prop_binSR_muon2016M_bin10
27 prop_binCRWS_electron2018_bin0
28 prop_binSR_muon2017_bin6
29 prop_binSR_electron2018_bin9
30 prop_binSR_muon2017_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

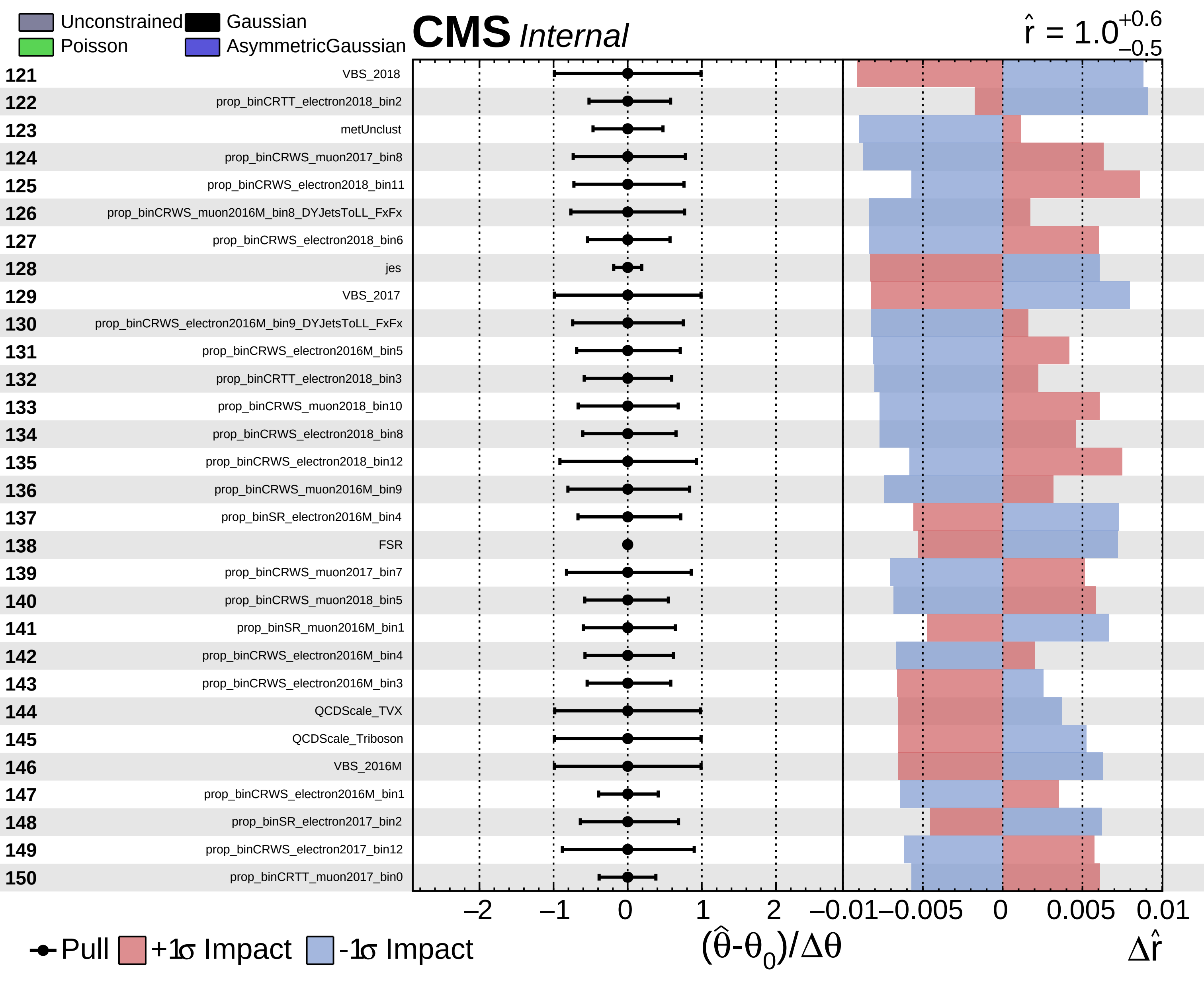

CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
31 prop_binCRTT_muon2018_bin0
32 prop_binSR_electron2016M_bin11
33 prop_binCRWS_electron2018_bin2
34 prop_binSR_electron2018_bin6
35 PF
36 prop_binCRWS_muon2018_bin3
37 prop_binSR_electron2017_bin10
38 prop_binSR_muon2017_bin8
39 prop_binCRWS_muon2017_bin4
40 lumi_2018
41 prop_binSR_muon2016M_bin6
42 prop_binSR_muon2016M_bin9
43 prop_binSR_muon2018_bin4
44 prop_binSR_electron2018_bin8
45 prop_binSR_muon2017_bin5
46 prop_binCRWS_electron2018_bin7
47 prop_binSR_muon2017_bin7
48 pdf_total_DY
49 prop_binSR_electron2017_bin6
50 lumi_2017
51 prop_binSR_muon2017_bin4
52 prop_binSR_muon2018_bin3
53 prop_binCRWS_muon2017_bin2
54 prop_binSR_electron2018_bin5
55 prop_binCRWS_electron2017_bin4
56 prop_binSR_muon2016M_bin5
57 prop_binSR_electron2016M_bin10
58 prop_binSR_electron2018_bin4
59 prop_binSR_electron2018_bin7
60 prop_binSR_muon2016M_bin8
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



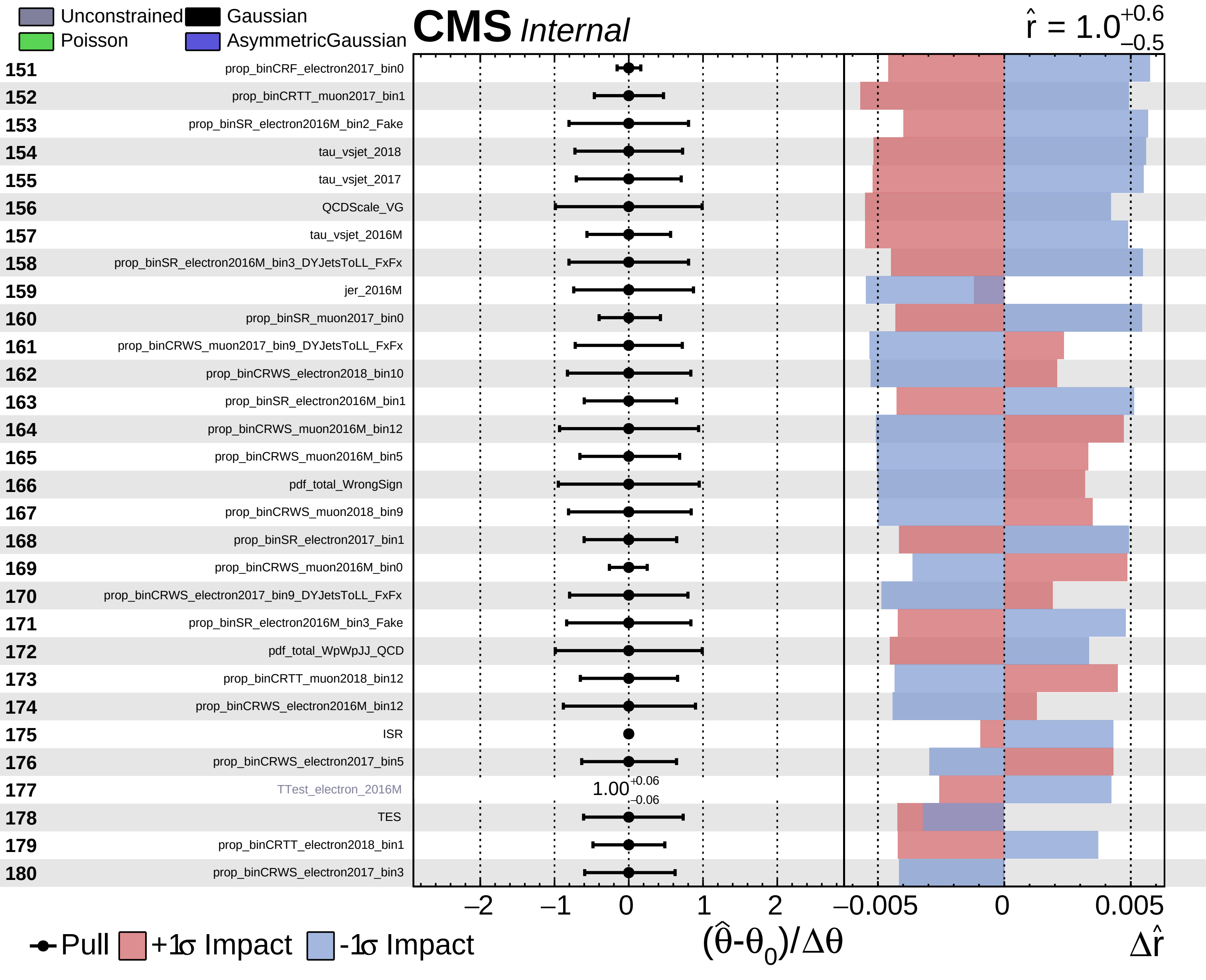
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
61 prop_binSR_electron2017_bin9
62 prop_binSR_muon2017_bin3
63 prop_binSR_electron2016M_bin9
64 prop_binSR_muon2018_bin2
65 prop_binSR_electron2017_bin5
66 prop_binCRWS_muon2016M_bin4
67 prop_binSR_electron2016M_bin6
68 prop_binSR_muon2016M_bin4
69 prop_binSR_muon2016M_bin7
70 lumi_2016M
71 prop_binCRWS_electron2016M_bin0
72 prop_binSR_muon2016M_bin3
73 prop_binCRWS_muon2018_bin8
74 prop_binSR_electron2016M_bin5
75 prop_binSR_muon2017_bin2
76 prop_binSR_electron2017_bin8
77 prop_binCRWS_muon2016M_bin7
78 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
79 prop_binCRWS_muon2017_bin3
80 prop_binCRWS_muon2016M_bin6
81 QCDScale_WpWpJJ_QCD
82 prop_binSR_electron2017_bin7
83 prop_binSR_electron2016M_bin8
84 prop_binCRWS_muon2018_bin1
85 prop_binSR_muon2018_bin1
86 pu
87 prop_binCRWS_electron2017_bin8
88 prop_binSR_electron2017_bin4
89 prop_binCRWS_muon2017_bin6
90 prop_binSR_electron2018_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

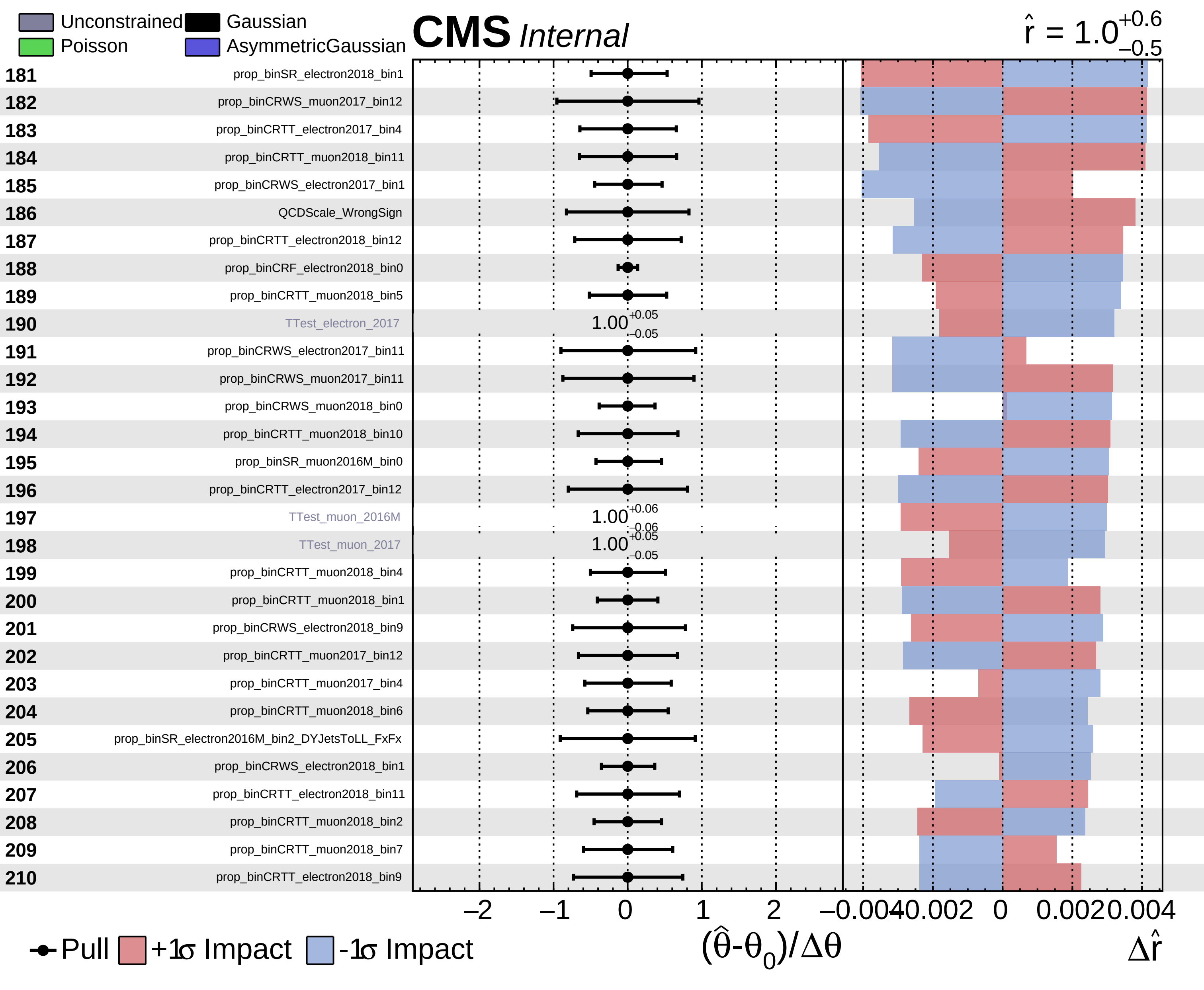


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
91 prop_binCRWS_muon2016M_bin1
92 prop_binSR_muon2016M_bin2
93 prop_binCRWS_electron2017_bin2
94 jer_2018
95 QCDScale_TTdilep
96 prop_binCRF_muon2016M_bin0
97 prop_binSR_electron2016M_bin7
98 prop_binCRWS_muon2017_bin1
99 btag
100 prop_binCRWS_electron2017_bin10
101 prop_binCRWS_muon2018_bin7
102 prop_binCRTT_electron2017_bin1
103 prop_binCRWS_muon2018_bin12
104 prop_binCRTT_electron2017_bin0
105 mistag
106 prop_binCRWS_muon2018_bin11
107 prop_binCRWS_muon2016M_bin3
108 prop_binCRTT_electron2016M_bin0
109 prop_binCRF_muon2017_bin0
110 prop_binSR_muon2017_bin1
111 prop_binCRF_muon2018_bin0
112 prop_binCRWS_electron2017_bin0
113 prop_binCRWS_muon2018_bin6
114 prop_binSR_electron2018_bin2
115 prop_binSR_electron2017_bin3
116 prop_binCRWS_muon2018_bin2
117 prop_binCRWS_electron2018_bin3
118 prop_binCRWS_electron2016M_bin2
119 prop_binCRTT_electron2018_bin0
120 prop_binCRWS_muon2017_bin5
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

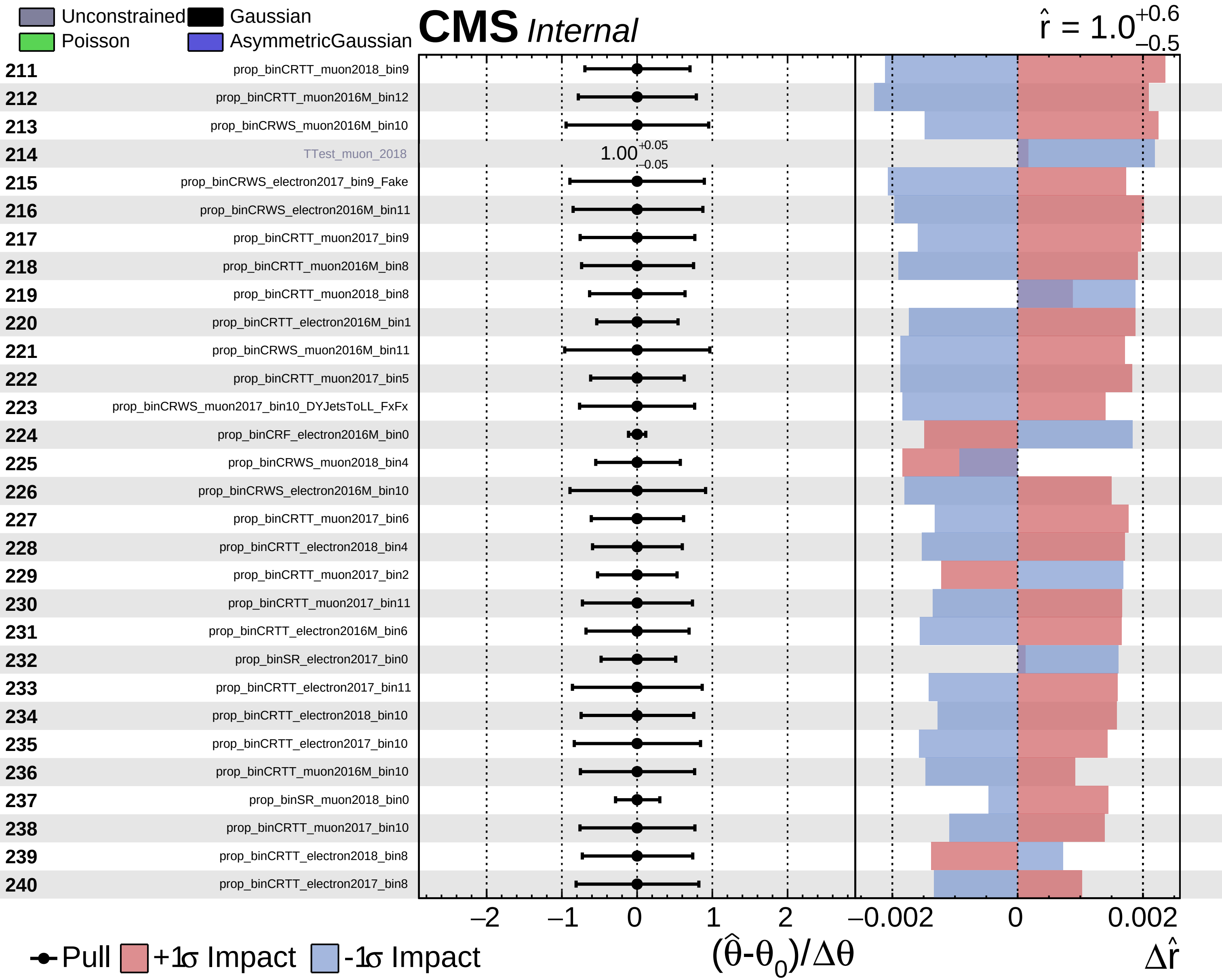


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
121 VBS_2018
122 prop_binCRTT_electron2018_bin2
123 metUnclust
124 prop_binCRWS_muon2017_bin8
125 prop_binCRWS_electron2018_bin11
126 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
127 prop_binCRWS_electron2018_bin6
128 jes
129 VBS_2017
130 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
131 prop_binCRWS_electron2016M_bin5
132 prop_binCRTT_electron2018_bin3
133 prop_binCRWS_muon2018_bin10
134 prop_binCRWS_electron2018_bin8
135 prop_binCRWS_electron2018_bin12
136 prop_binCRWS_muon2016M_bin9
137 prop_binSR_electron2016M_bin4
138 FSR
139 prop_binCRWS_muon2017_bin7
140 prop_binCRWS_muon2018_bin5
141 prop_binSR_muon2016M_bin1
142 prop_binCRWS_electron2016M_bin4
143 prop_binCRWS_electron2016M_bin3
144 QCDScale_TVX
145 QCDScale_Triboson
146 VBS_2016M
147 prop_binCRWS_electron2016M_bin1
148 prop_binSR_electron2017_bin2
149 prop_binCRWS_electron2017_bin12
150 prop_binCRTT_muon2017_bin0
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

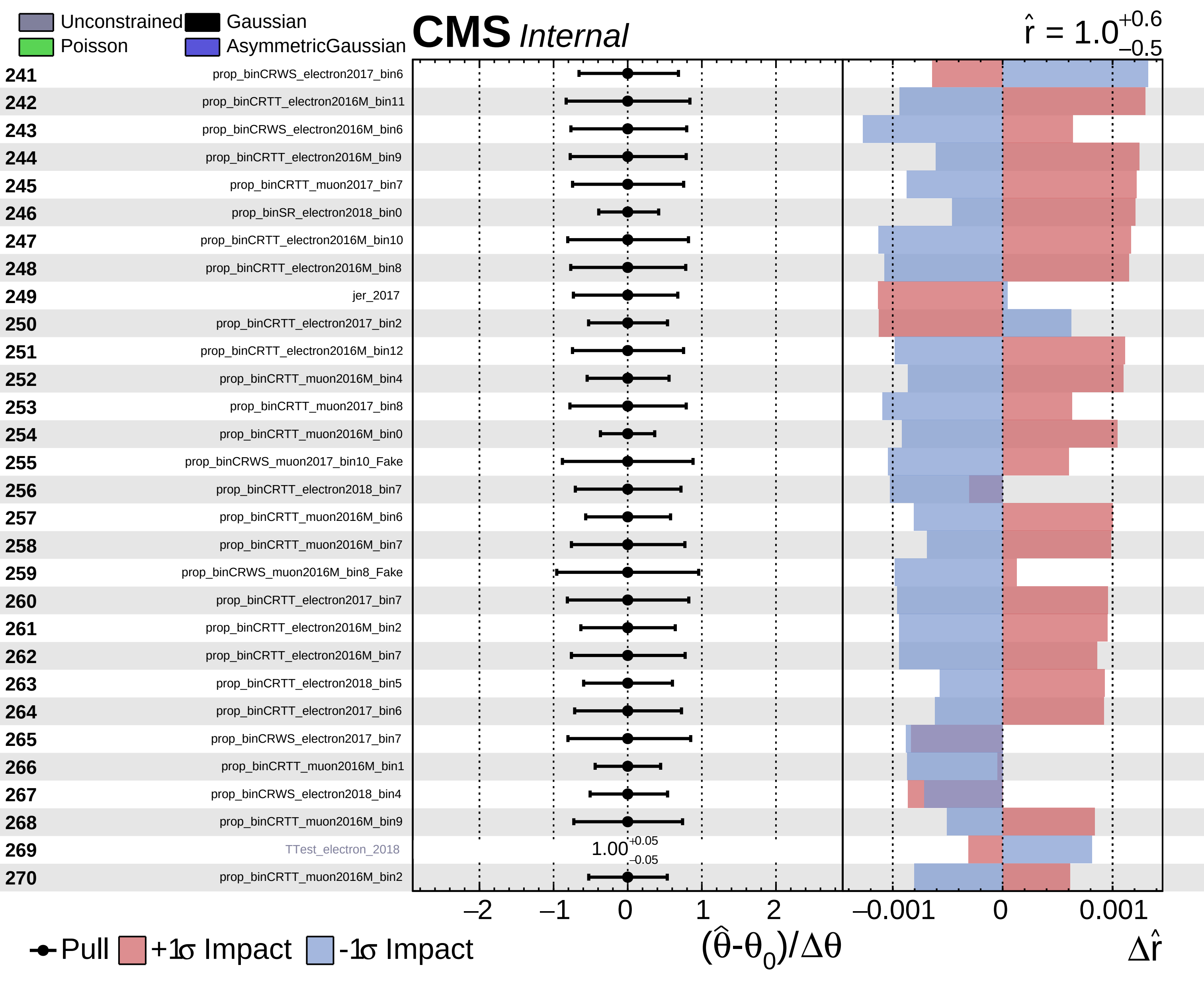

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
151 prop_binCRF_electron2017_bin0
152 prop_binCRTT_muon2017_bin1
153 prop_binSR_electron2016M_bin2_Fake
154 tau_vsjet_2018
155 tau_vsjet_2017
156 QCDScale_VG
157 tau_vsjet_2016M
158 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
159 jer_2016M
160 prop_binSR_muon2017_bin0
161 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
162 prop_binCRWS_electron2018_bin10
163 prop_binSR_electron2016M_bin1
164 prop_binCRWS_muon2016M_bin12
165 prop_binCRWS_muon2016M_bin5
166 pdf_total_WrongSign
167 prop_binCRWS_muon2018_bin9
168 prop_binSR_electron2017_bin1
169 prop_binCRWS_muon2016M_bin0
170 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
171 prop_binSR_electron2016M_bin3_Fake
172 pdf_total_WpWpJJ_QCD
173 prop_binCRTT_muon2018_bin12
174 prop_binCRWS_electron2016M_bin12
175 ISR
176 prop_binCRWS_electron2017_bin5
177 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
178 TES
179 prop_binCRTT_electron2018_bin1
180 prop_binCRWS_electron2017_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



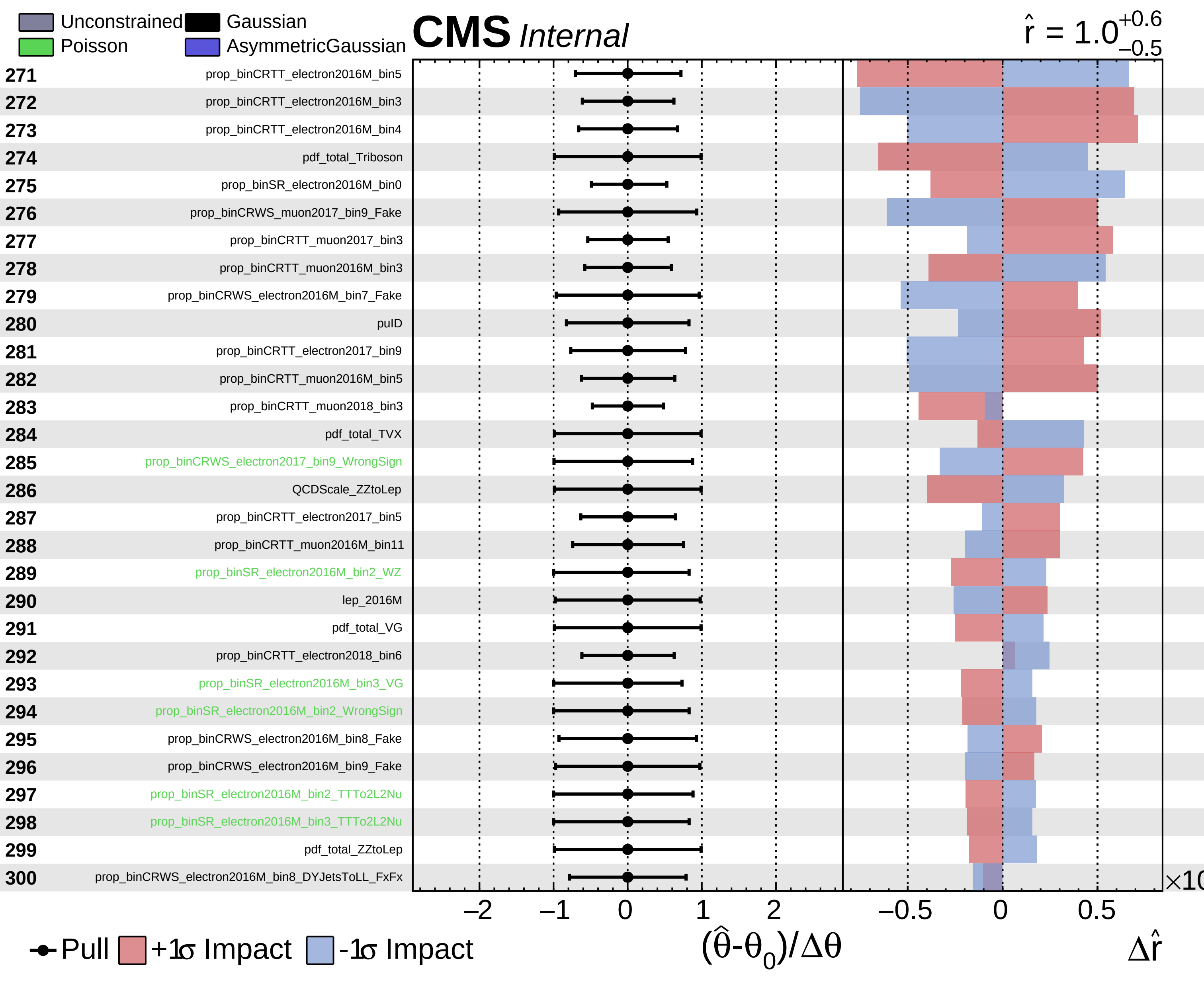
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
181 prop_binSR_electron2018_bin1
182 prop_binCRWS_muon2017_bin12
183 prop_binCRTT_electron2017_bin4
184 prop_binCRTT_muon2018_bin11
185 prop_binCRWS_electron2017_bin1
186 QCDScale_WrongSign
187 prop_binCRTT_electron2018_bin12
188 prop_binCRF_electron2018_bin0
189 prop_binCRTT_muon2018_bin5
190 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
191 prop_binCRWS_electron2017_bin11
192 prop_binCRWS_muon2017_bin11
193 prop_binCRWS_muon2018_bin0
194 prop_binCRTT_muon2018_bin10
195 prop_binSR_muon2016M_bin0
196 prop_binCRTT_electron2017_bin12
197 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
198 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
199 prop_binCRTT_muon2018_bin4
200 prop_binCRTT_muon2018_bin1
201 prop_binCRWS_electron2018_bin9
202 prop_binCRTT_muon2017_bin12
203 prop_binCRTT_muon2017_bin4
204 prop_binCRTT_muon2018_bin6
205 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
206 prop_binCRWS_electron2018_bin1
207 prop_binCRTT_electron2018_bin11
208 prop_binCRTT_muon2018_bin2
209 prop_binCRTT_muon2018_bin7
210 prop_binCRTT_electron2018_bin9
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



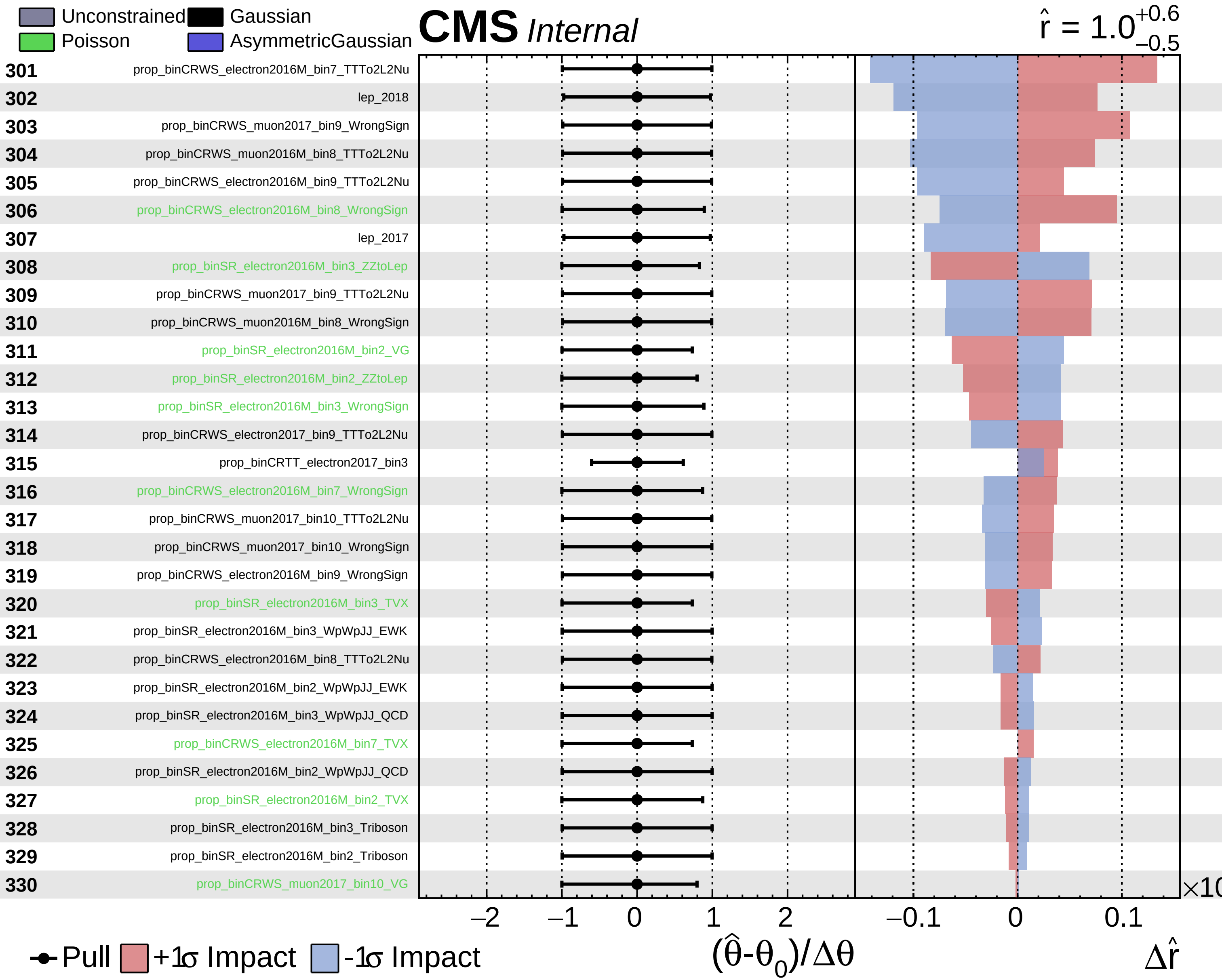
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
211 prop_binCRTT_muon2018_bin9
212 prop_binCRTT_muon2016M_bin12
213 prop_binCRWS_muon2016M_bin10
214 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
215 prop_binCRWS_electron2017_bin9_Fake
216 prop_binCRWS_electron2016M_bin11
217 prop_binCRTT_muon2017_bin9
218 prop_binCRTT_muon2016M_bin8
219 prop_binCRTT_muon2018_bin8
220 prop_binCRTT_electron2016M_bin1
221 prop_binCRWS_muon2016M_bin11
222 prop_binCRTT_muon2017_bin5
223 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
224 prop_binCRF_electron2016M_bin0
225 prop_binCRWS_muon2018_bin4
226 prop_binCRWS_electron2016M_bin10
227 prop_binCRTT_muon2017_bin6
228 prop_binCRTT_electron2018_bin4
229 prop_binCRTT_muon2017_bin2
230 prop_binCRTT_muon2017_bin11
231 prop_binCRTT_electron2016M_bin6
232 prop_binSR_electron2017_bin0
233 prop_binCRTT_electron2017_bin11
234 prop_binCRTT_electron2018_bin10
235 prop_binCRTT_electron2017_bin10
236 prop_binCRTT_muon2016M_bin10
237 prop_binSR_muon2018_bin0
238 prop_binCRTT_muon2017_bin10
239 prop_binCRTT_electron2018_bin8
240 prop_binCRTT_electron2017_bin8
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



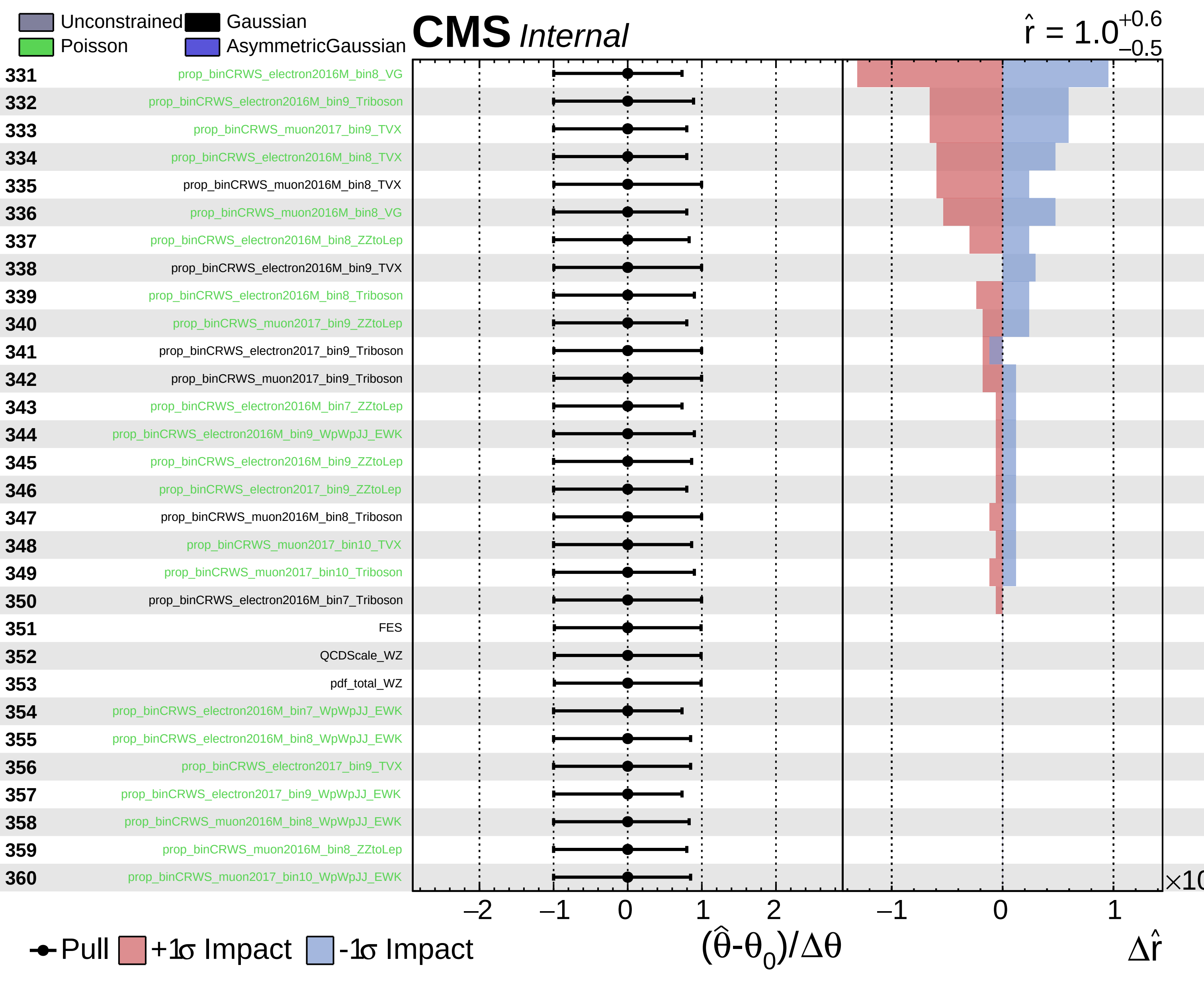
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
241 prop_binCRWS_electron2017_bin6
242 prop_binCRTT_electron2016M_bin11
243 prop_binCRWS_electron2016M_bin6
244 prop_binCRTT_electron2016M_bin9
245 prop_binCRTT_muon2017_bin7
246 prop_binSR_electron2018_bin0
247 prop_binCRTT_electron2016M_bin10
248 prop_binCRTT_electron2016M_bin8
249 jer_2017
250 prop_binCRTT_electron2017_bin2
251 prop_binCRTT_electron2016M_bin12
252 prop_binCRTT_muon2016M_bin4
253 prop_binCRTT_muon2017_bin8
254 prop_binCRTT_muon2016M_bin0
255 prop_binCRWS_muon2017_bin10_Fake
256 prop_binCRTT_electron2018_bin7
257 prop_binCRTT_muon2016M_bin6
258 prop_binCRTT_muon2016M_bin7
259 prop_binCRWS_muon2016M_bin8_Fake
260 prop_binCRTT_electron2017_bin7
261 prop_binCRTT_electron2016M_bin2
262 prop_binCRTT_electron2016M_bin7
263 prop_binCRTT_electron2018_bin5
264 prop_binCRTT_electron2017_bin6
265 prop_binCRWS_electron2017_bin7
266 prop_binCRTT_muon2016M_bin1
267 prop_binCRWS_electron2018_bin4
268 prop_binCRTT_muon2016M_bin9
269 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
270 prop_binCRTT_muon2016M_bin2
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



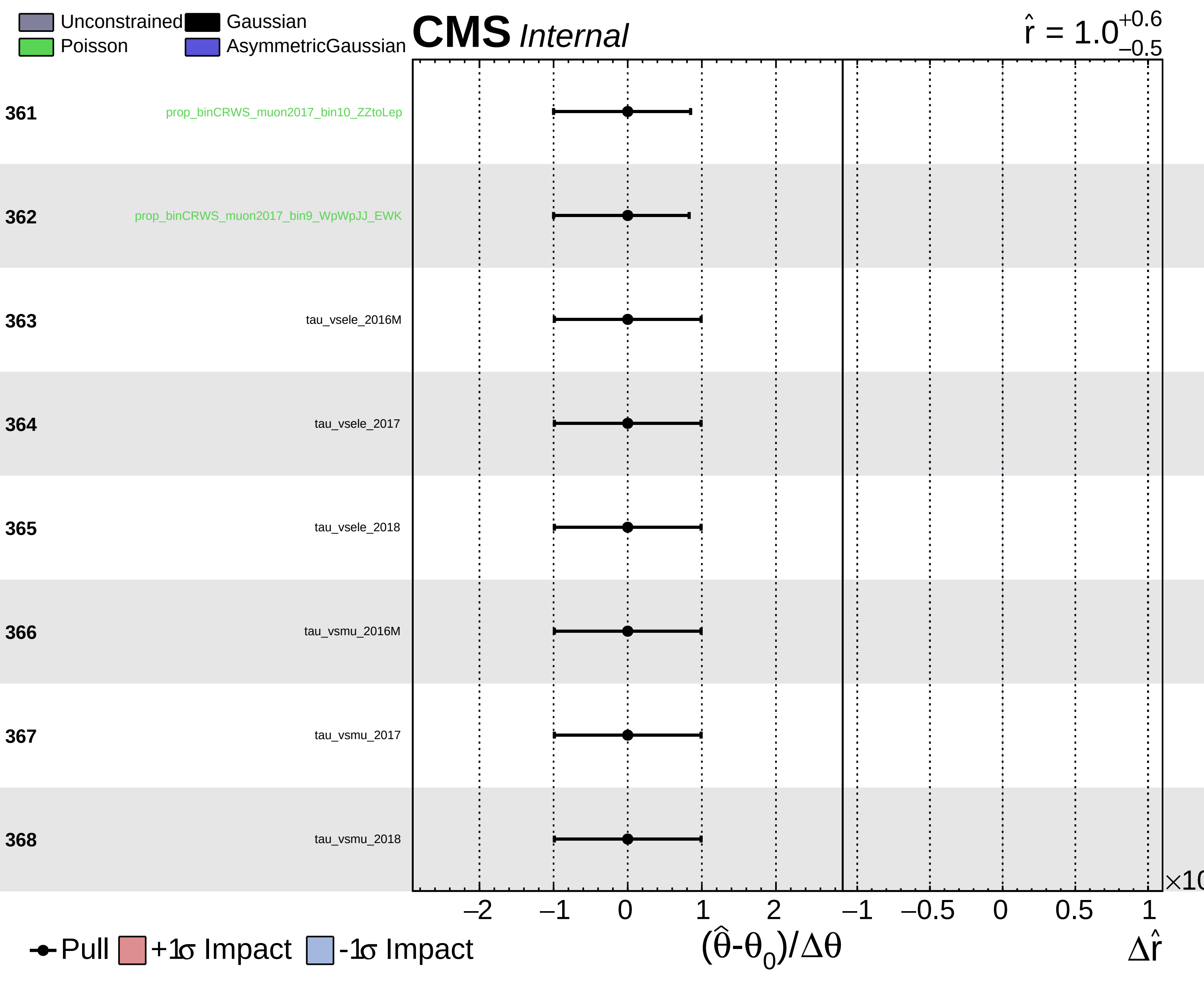
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
271 prop_binCRTT_electron2016M_bin5
272 prop_binCRTT_electron2016M_bin3
273 prop_binCRTT_electron2016M_bin4
274 pdf_total_Triboson
275 prop_binSR_electron2016M_bin0
276 prop_binCRWS_muon2017_bin9_Fake
277 prop_binCRTT_muon2017_bin3
278 prop_binCRTT_muon2016M_bin3
279 prop_binCRWS_electron2016M_bin7_Fake
280 puID
281 prop_binCRTT_electron2017_bin9
282 prop_binCRTT_muon2016M_bin5
283 prop_binCRTT_muon2018_bin3
284 pdf_total_TVX
285 prop_binCRWS_electron2017_bin9_WrongSign
286 QCDScale_ZZtoLep
287 prop_binCRTT_electron2017_bin5
288 prop_binCRTT_muon2016M_bin11
289 prop_binSR_electron2016M_bin2_WZ
290 lep_2016M
291 pdf_total_VG
292 prop_binCRTT_electron2018_bin6
293 prop_binSR_electron2016M_bin3_VG
294 prop_binSR_electron2016M_bin2_WrongSign
295 prop_binCRWS_electron2016M_bin8_Fake
296 prop_binCRWS_electron2016M_bin9_Fake
297 prop_binSR_electron2016M_bin2_TTTo2L2Nu
298 prop_binSR_electron2016M_bin3_TTTo2L2Nu
299 pdf_total_ZZtoLep
300 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
301 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
302 lep_2018
303 prop_binCRWS_muon2017_bin9_WrongSign
304 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
305 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
306 prop_binCRWS_electron2016M_bin8_WrongSign
307 lep_2017
308 prop_binSR_electron2016M_bin3_ZZtoLep
309 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
310 prop_binCRWS_muon2016M_bin8_WrongSign
311 prop_binSR_electron2016M_bin2_VG
312 prop_binSR_electron2016M_bin2_ZZtoLep
313 prop_binSR_electron2016M_bin3_WrongSign
314 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
315 prop_binCRTT_electron2017_bin3
316 prop_binCRWS_electron2016M_bin7_WrongSign
317 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
318 prop_binCRWS_muon2017_bin10_WrongSign
319 prop_binCRWS_electron2016M_bin9_WrongSign
320 prop_binSR_electron2016M_bin3_TVX
321 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
322 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
323 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
324 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
325 prop_binCRWS_electron2016M_bin7_TVX
326 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
327 prop_binSR_electron2016M_bin2_TVX
328 prop_binSR_electron2016M_bin3_Triboson
329 prop_binSR_electron2016M_bin2_Triboson
330 prop_binCRWS_muon2017_bin10_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
331 prop_binCRWS_electron2016M_bin8_VG
332 prop_binCRWS_electron2016M_bin9_Triboson
333 prop_binCRWS_muon2017_bin9_TVX
334 prop_binCRWS_electron2016M_bin8_TVX
335 prop_binCRWS_muon2016M_bin8_TVX
336 prop_binCRWS_muon2016M_bin8_VG
337 prop_binCRWS_electron2016M_bin8_ZZtoLep
338 prop_binCRWS_electron2016M_bin9_TVX
339 prop_binCRWS_electron2016M_bin8_Triboson
340 prop_binCRWS_muon2017_bin9_ZZtoLep
341 prop_binCRWS_electron2017_bin9_Triboson
342 prop_binCRWS_muon2017_bin9_Triboson
343 prop_binCRWS_electron2016M_bin7_ZZtoLep
344 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
345 prop_binCRWS_electron2016M_bin9_ZZtoLep
346 prop_binCRWS_electron2017_bin9_ZZtoLep
347 prop_binCRWS_muon2016M_bin8_Triboson
348 prop_binCRWS_muon2017_bin10_TVX
349 prop_binCRWS_muon2017_bin10_Triboson
350 prop_binCRWS_electron2016M_bin7_Triboson
351 FES
352 QCDScale_WZ
353 pdf_total_WZ
354 prop_binCRWS_electron2016M_bin7_WpWpJJ_EWK
355 prop_binCRWS_electron2016M_bin8_WpWpJJ_EWK
356 prop_binCRWS_electron2017_bin9_TVX
357 prop_binCRWS_electron2017_bin9_WpWpJJ_EWK
358 prop_binCRWS_muon2016M_bin8_WpWpJJ_EWK
359 prop_binCRWS_muon2016M_bin8_ZZtoLep
360 prop_binCRWS_muon2017_bin10_WpWpJJ_EWK
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
361
prop_binCRWS_muon2017_bin10_ZZtoLep
362
prop_binCRWS_muon2017_bin9_WpWpJJ_EWK
363
tau_vsele_2016M
364
tau_vsele_2017
365
tau_vsele_2018
366
tau_vsmu_2016M
367
tau_vsmu_2017
368
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$